SYSTEM DOCUMENTATION

CARD MRI Rizal Bank Leave Approval & Routing System (LAR-Sys)

Course: Workplace Systems Development	**Batch / Section:** CMS BSAIS Batch 1, Block 1
Institution: CARD MRI Rizal Bank Inc.	**Group Members:** Arines, J., Apelado, A., Bagotayao, A., Cruz, M., Dela Cruz, M., Mercene, J.

1. SYSTEM TITLE, DESCRIPTION, AND OBJECTIVES

System Title: CARD MRI Rizal Bank Leave Approval & Routing System (LAR-Sys)

System Description: LAR-Sys is an enterprise web application designed for CARD MRI Rizal Bank Inc. to streamline leave management across field and branch staff. It corrects the flawed routing mechanism of the legacy PeopleCore portal by re-establishing proper supervisory hierarchy, incorporating mandatory reliever assignment, and automating multi-tier approval flows for Account Officers, Unit Managers, and Area Managers.

System Objectives:

- **Re-establish Supervisory Oversight:** Automatically route Account Officer leave requests to their immediate Unit Manager (UM) before escalation.
- **Ensure Operational Continuity:** Enforce mandatory selection of branch relievers during leave filing to prevent unassigned center meetings and uncollected microfinance loans.
- **Reduce Administrative Bottlenecks:** Provide localized approval authority to Unit Managers to eliminate delays caused by high-level managerial backlogs.
- **Provide Live Audit Trail:** Maintain real-time tracking dashboards and immutable historical records for HR balance adjustments.

2. BUSINESS PROCESS

The core business process addresses leave filing and workload handover within microfinance operations:

1. **Leave Request & Reliever Assignment:** An Account Officer (AO) initiates a leave application via the self-service portal, selecting inclusive dates, leave type, and assigning an available branch reliever.
2. **Level-1 Supervisory Review:** The system routes the application to the immediate Unit Manager (UM). The UM checks local unit staffing, center meeting schedules, and reliever availability before endorsing the request.
3. **Level-2 Final Approval:** Upon UM endorsement, the request is forwarded to the Area Manager (AM) for final cluster sign-off.
4. **HRIS Log & Notification:** Approved requests automatically update employee leave balances, notify all stakeholders, and log audit history.

3. ENTITY RELATIONSHIP DIAGRAM (ERD)

The LAR-Sys relational database structure consists of three core entities ensuring complete relational integrity between employees, filed applications, and sequential management approvals:

Entity Name	Keys & Foreign Keys	Attributes & Data Types
EMPLOYEE	PK: `employee_id` (INT)	`full_name` (VARCHAR), `role` (ENUM: AO, UM, AM), `unit_branch` (VARCHAR), `leave_balance` (INT)
LEAVE_APPLICATION	PK: `leave_id` (INT) FK: `employee_id` (INT)	`leave_type` (VARCHAR), `start_date` (DATE), `end_date` (DATE), `reliever_name` (VARCHAR), `overall_status` (VARCHAR)
APPROVAL_WORKFLOW	PK: `workflow_id` (INT) FK: `leave_id` (INT)	`um_recommendation` (ENUM), `am_approval` (ENUM), `remarks` (TEXT), `timestamp` (DATETIME)

4. DATA FLOW DIAGRAM (DFD)

Level-0 Context Data Flow Diagram:

```
+--------------------+ Leave Application Data +-------------------------+ | | ---------
---------------------------------> | | | Account Officer | | 0.0 LAR-Sys Core Engine| |
(AO) | <---------------------------------------- | | +--------------------+ Status
Notifications +-------------------------+ | ^ Forward | | UM/AM Request v | Decision +-
-------------------------+ | Unit Manager (UM) / | | Area Manager (AM) | +--------------
------------+
```

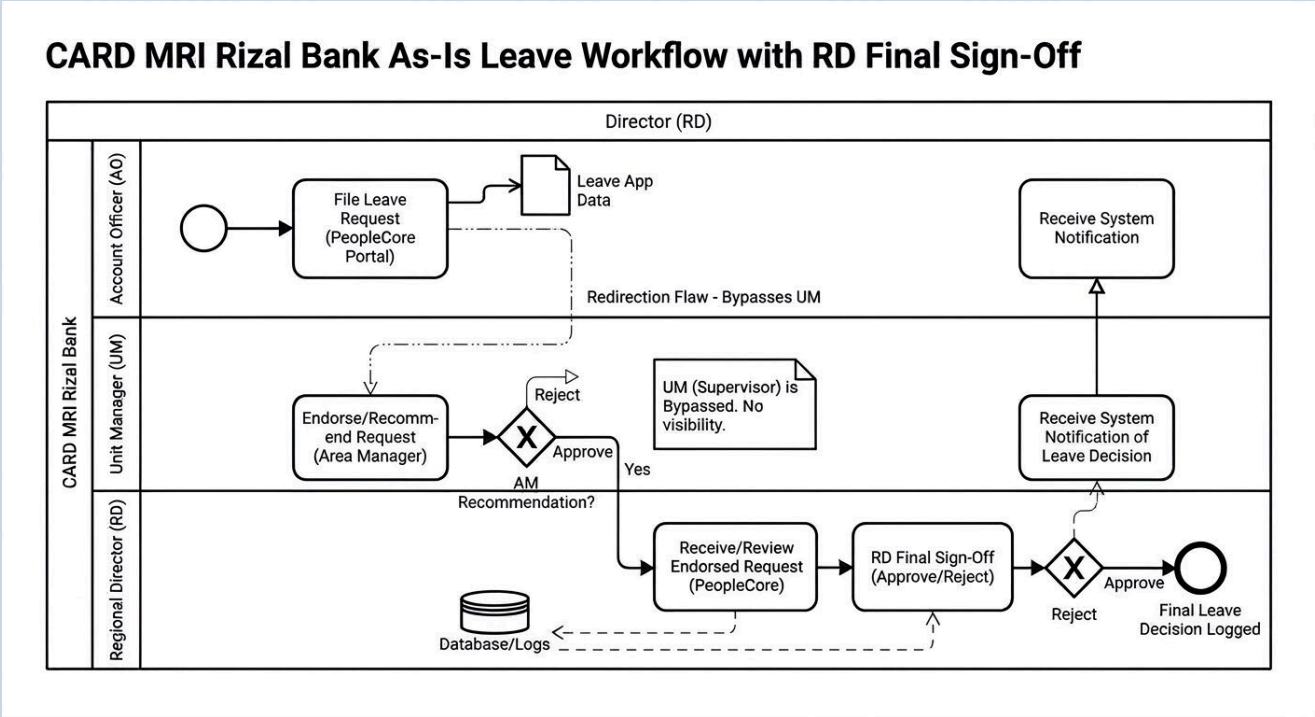
Level-1 Data Flow Process:

- **Process 1.0 (File Leave):** AO inputs application and reliever info → Stores in `LEAVE_APPLICATION` table.
- **Process 2.0 (UM Review):** System checks `LEAVE_APPLICATION` → Fetches UM schedule → Writes `um_recommendation` to `APPROVAL_WORKFLOW`.
- **Process 3.0 (AM Approval):** System filters UM-endorsed applications → Writes `am_approval` → Triggers `EMPLOYEE` balance update.

5. BUSINESS PROCESS MODEL AND NOTATION (BPMN 2.0)

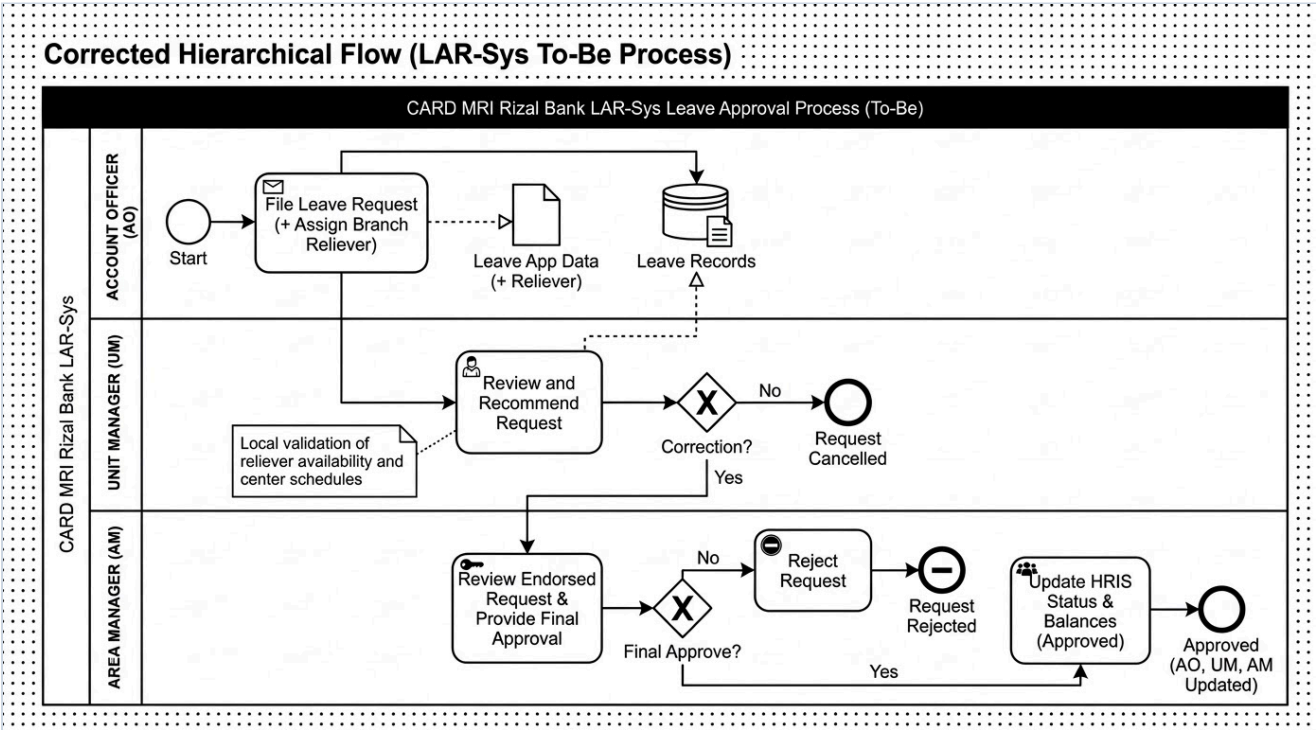
As-Is BPMN Process Flow (Flawed Hierarchy)

In the existing PeopleCore workflow, filed leave applications bypass the immediate Unit Manager (UM) and route directly to the Area Manager (AM), causing scheduling conflicts and unmonitored field operations.



To-Be BPMN Process Flow (Corrected Hierarchy)

The corrected LAR-Sys process routes filings first to the Unit Manager for Level-1 recommendation and reliever validation, followed by Area Manager final approval and automated HRIS updates.



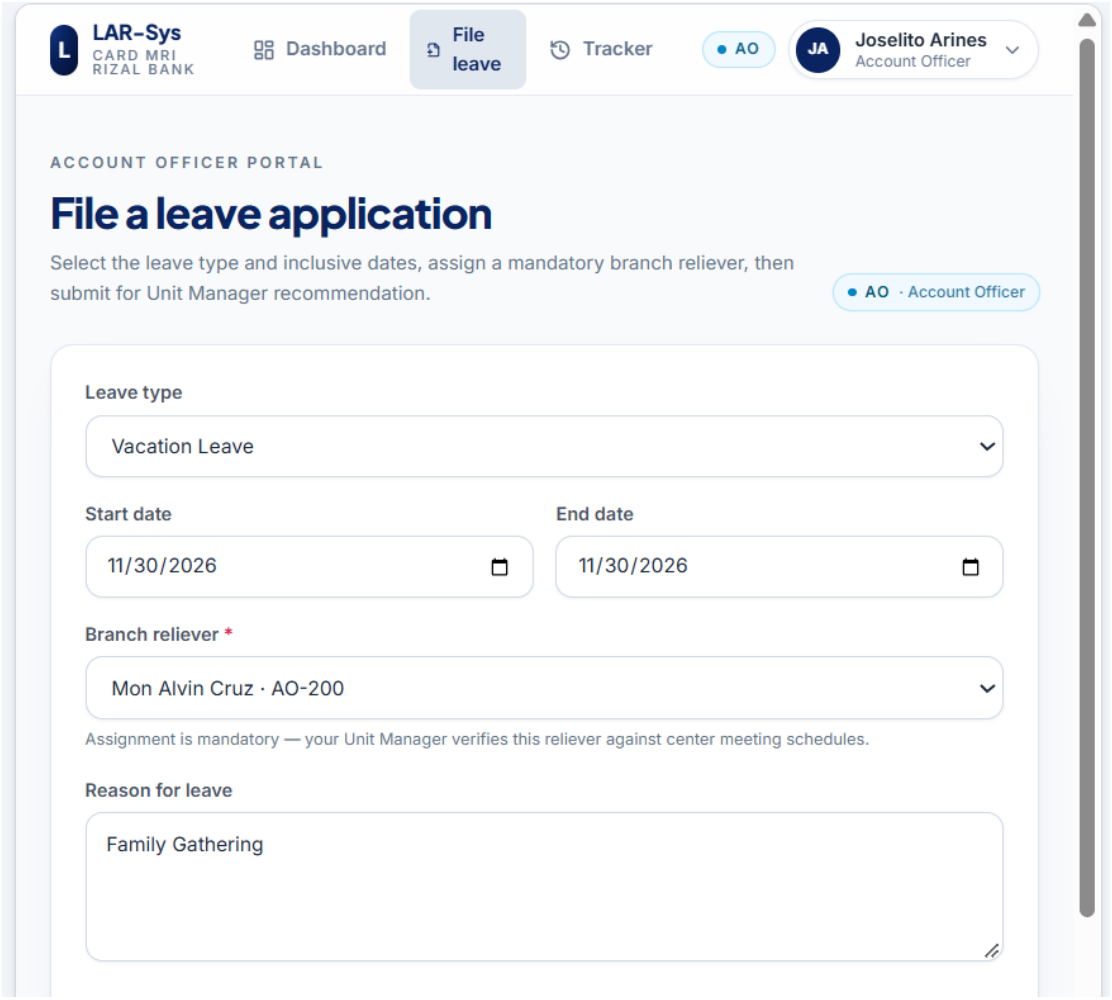
6. SYSTEM FEATURES AND FUNCTIONALITIES

- **FR-1 (Self-Service Leave Filing):** Enables field Account Officers to initiate leave requests with integrated leave balance counters and mandatory branch reliever selection.
- **FR-2 (Sequential Multi-Tier Routing):** Automatically enforces sequential Level-1 Unit Manager recommendation before releasing the file to Level-2 Area Manager final sign-off.
- **FR-3 (Real-Time Status Dashboard):** Provides transparent visual progress tracking (`Pending UM Review` → `Pending AM Approval` → `Approved / Updated`).
- **FR-4 (Audit Logging & Data Integrity):** Automatically calculates leave deductions upon final sign-off and maintains an immutable audit log for HR compliance.

7. SCREENSHOTS OF THE DEVELOPED SYSTEM

Below are representative interface views from the functional LAR-Sys base prototype implementation:

System View / Module	Interface Component & Workflow Context
AO Self-Service Portal	Includes active leave balance overview, leave type drop-down, date selection, and compulsory *Branch Reliever* selection field.
UM Recommendation Panel	Dedicated manager queue listing pending unit leave requests, showing field coverage schedules, and recommendation buttons (Approve / Request Correction).
AM Final Approval Dashboard	Cluster-wide view for Area Managers to quickly inspect UM-endorsed requests and execute final cluster sign-offs.
Live Status Tracker	Real-time progress tracker visible to employees showing current workflow stage, timestamped approvals, and assigned reliever details.


LAR-Sys
CARD MRI
RIZAL BANK
Dashboard
File leave
Tracker
AO
JA
Joselito Arines
Account Officer
ACCOUNT OFFICER PORTAL
File a leave application
Select the leave type and inclusive dates, assign a mandatory branch reliever, then submit for Unit Manager recommendation.
AO · Account Officer
Leave type
Vacation Leave
Start date
11/30/2026
End date
11/30/2026
Branch reliever *
Mon Alvin Cruz · AO-200
Assignment is mandatory — your Unit Manager verifies this reliever against center meeting schedules.
Reason for leave
Family Gathering

REQUEST SUMMARY

Days requested	1
Available credits	11 day(s)
Balance after approval	10 day(s)

Nov 29, 2026 → Nov 29, 2026

RELIEVER CHECK

Mon Alvin Cruz is available
No overlapping leave on record for this period.

Field center meeting coverage still needs the Unit Manager's recommendation before the request moves to the Area Manager.

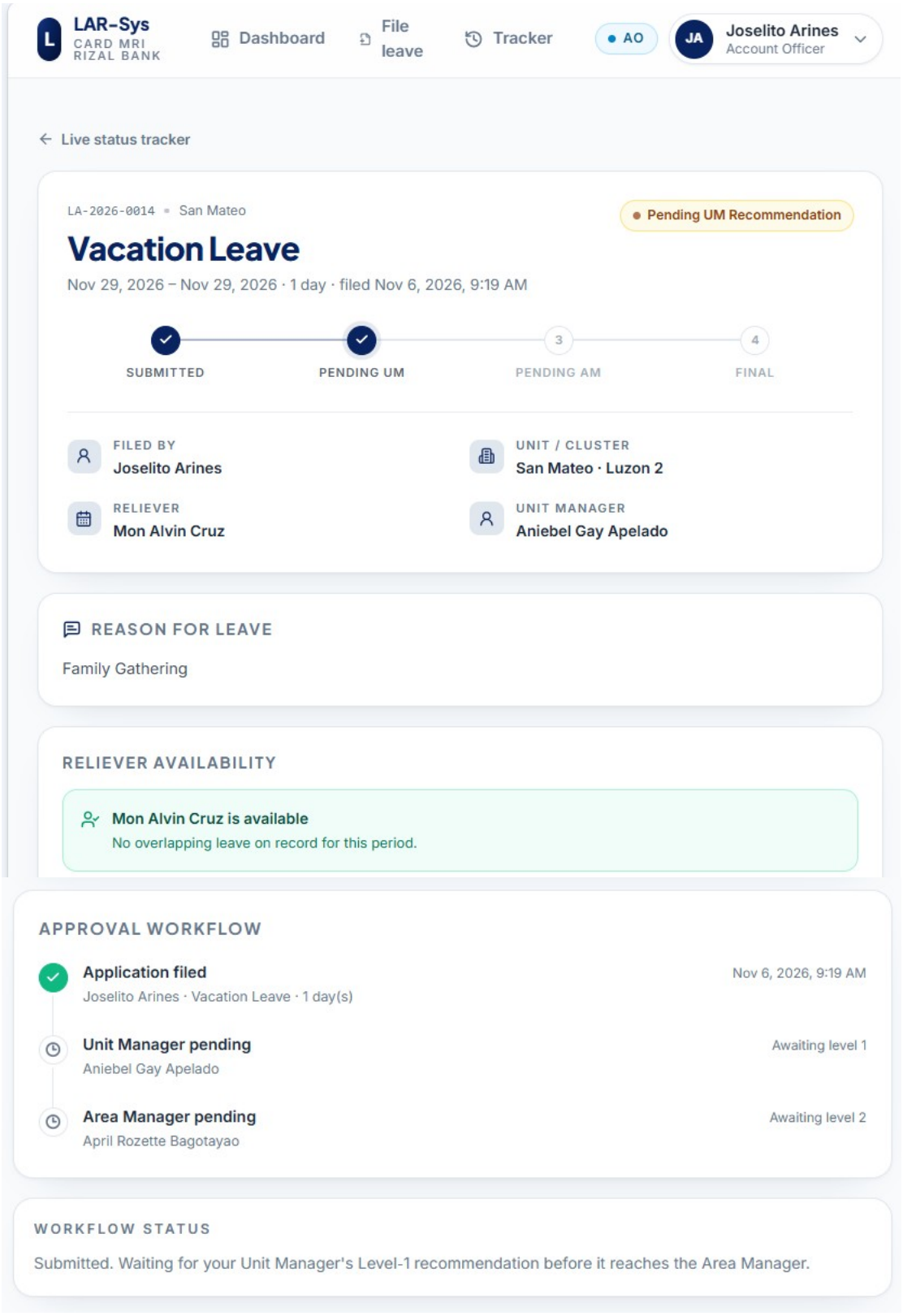
ROUTING PREVIEW

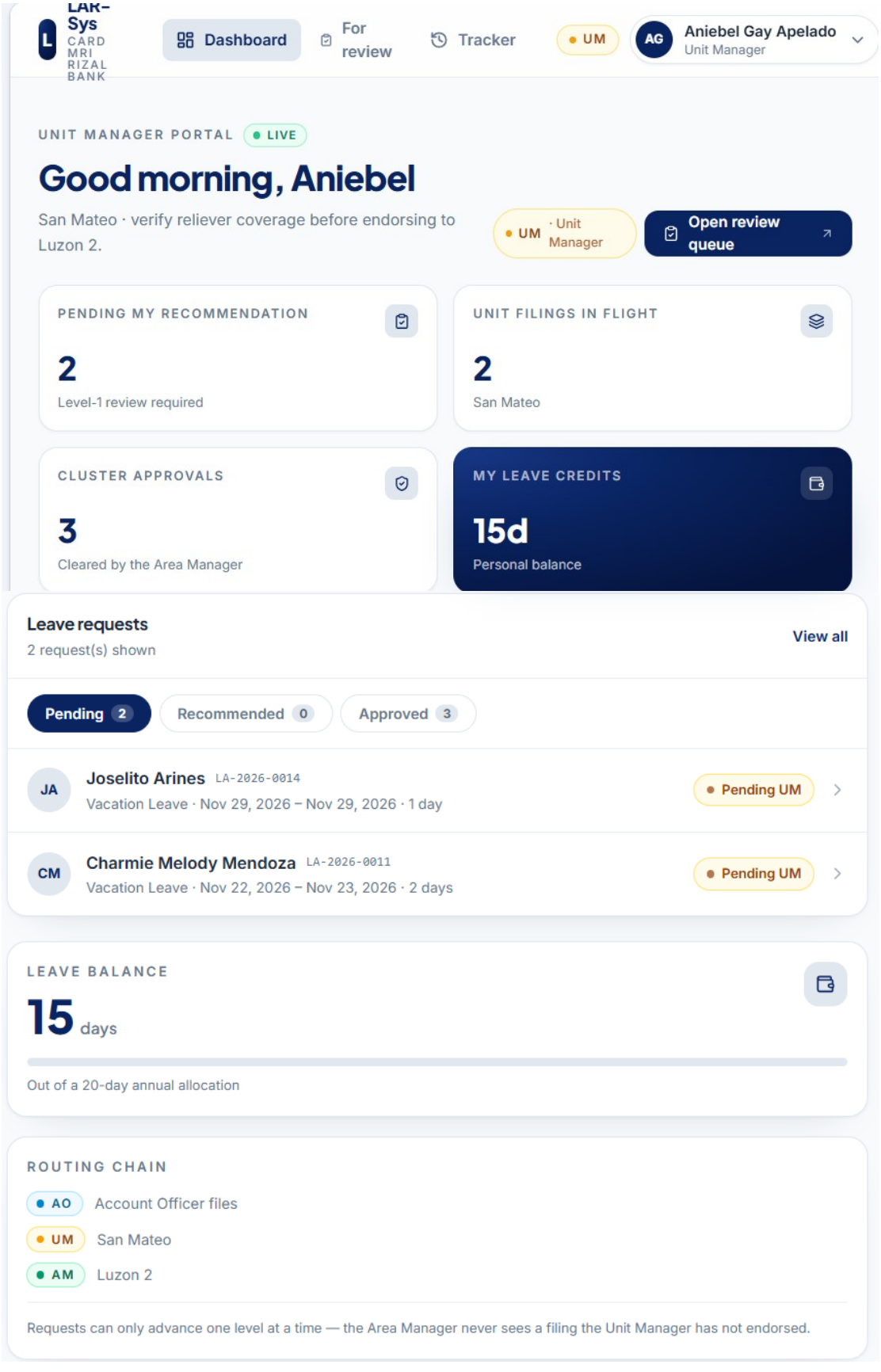
UM **Aniebel Gay Apelado**
Level-1 recommendation · San Mateo

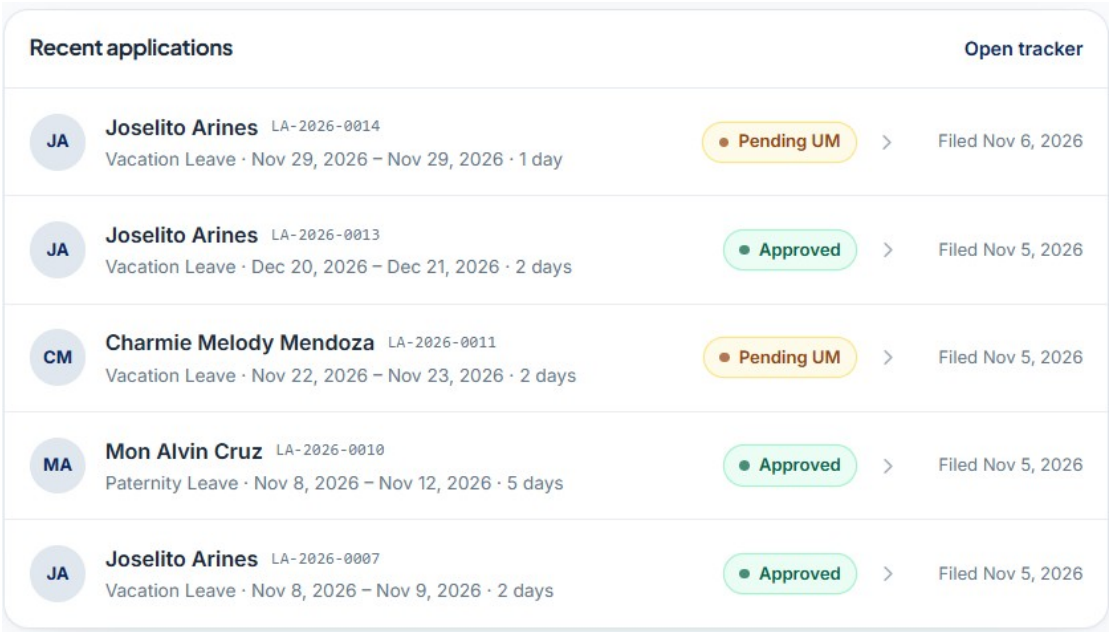
AM **April Rozette Bagotayao**
Level-2 final approval · Luzon 2

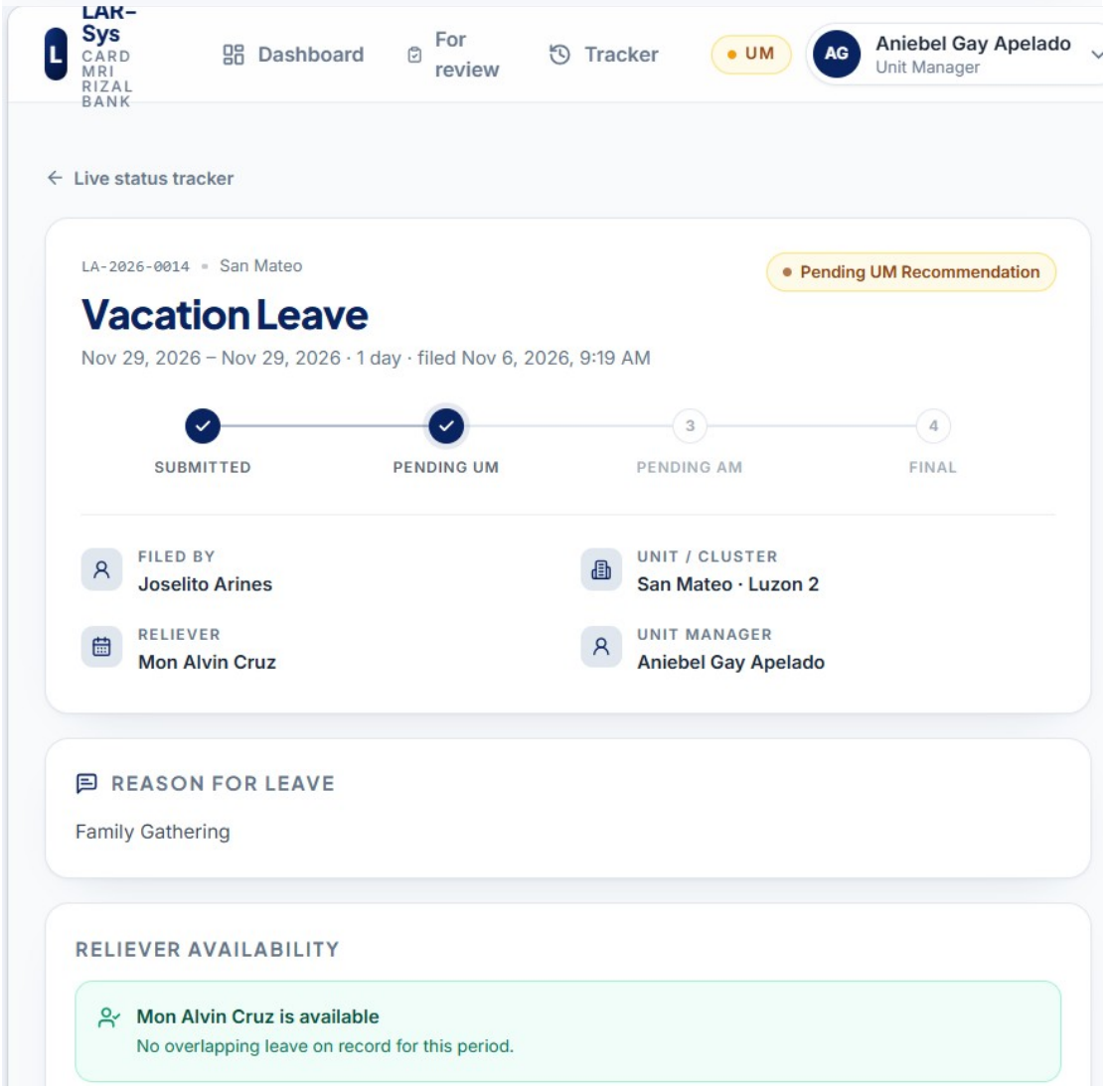
Sequential — no level can be skipped.

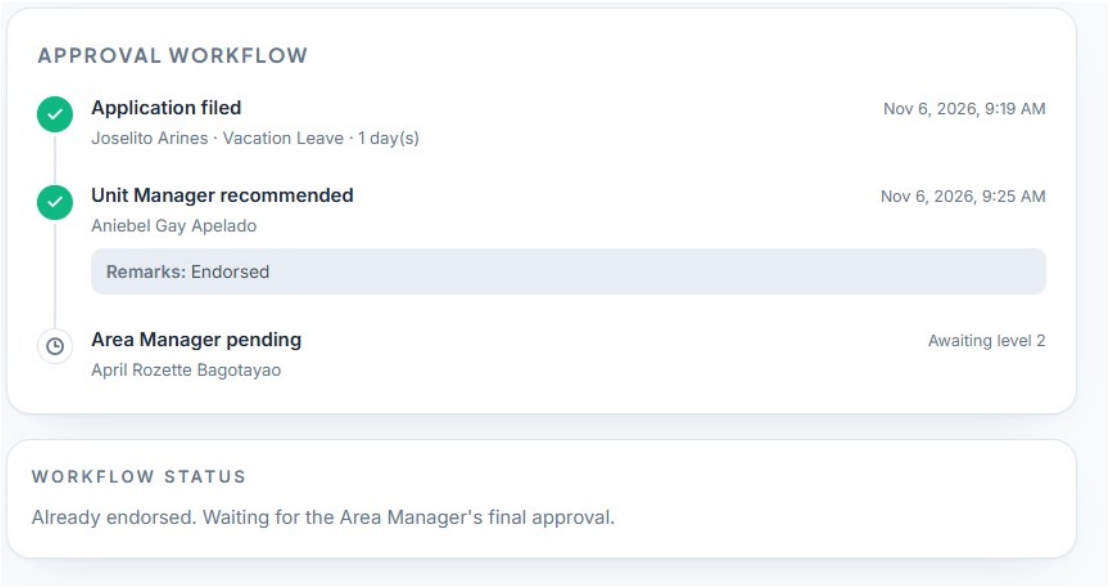
Account Officer · San Mateo


LAR-Sys
CARD MRI
RIZAL BANK
Dashboard
File leave
Tracker
AO
JA
Joselito Arines
Account Officer
Live status tracker
LA-2026-0014 • San Mateo
Pending UM Recommendation
Vacation Leave
Nov 29, 2026 – Nov 29, 2026 · 1 day · filed Nov 6, 2026, 9:19 AM
SUBMITTED
PENDING UM
3
PENDING AM
4
FINAL
FILED BY
Joselito Arines
UNIT / CLUSTER
San Mateo · Luzon 2
RELIEVER
Mon Alvin Cruz
UNIT MANAGER
Aniebel Gay Apelado
REASON FOR LEAVE
Family Gathering
RELIEVER AVAILABILITY
Mon Alvin Cruz is available
No overlapping leave on record for this period.
APPROVAL WORKFLOW
Application filed
Joselito Arines · Vacation Leave · 1 day(s)
Nov 6, 2026, 9:19 AM
Unit Manager pending
Aniebel Gay Apelado
Awaiting level 1
Area Manager pending
April Rozette Bagotayao
Awaiting level 2
WORKFLOW STATUS
Submitted. Waiting for your Unit Manager's Level-1 recommendation before it reaches the Area Manager.


LAR-Sys
CARD MRI RIZAL BANK
Dashboard
For review
Tracker
UM
AG
Aniebel Gay Apelado
Unit Manager
UNIT MANAGER PORTAL
LIVE
Good morning, Aniebel
San Mateo · verify reliever coverage before endorsing to Luzon 2.
UM · Unit Manager
Open review queue
PENDING MY RECOMMENDATION
2
Level-1 review required
UNIT FILINGS IN FLIGHT
2
San Mateo
CLUSTER APPROVALS
3
Cleared by the Area Manager
MY LEAVE CREDITS
15d
Personal balance
Leave requests
2 request(s) shown
View all
Pending 2
Recommended 0
Approved 3
JA
Joselito Arines
LA-2026-0014
Vacation Leave · Nov 29, 2026 – Nov 29, 2026 · 1 day
Pending UM
CM
Charmie Melody Mendoza
LA-2026-0011
Vacation Leave · Nov 22, 2026 – Nov 23, 2026 · 2 days
Pending UM
LEAVE BALANCE
15 days
Out of a 20-day annual allocation
ROUTING CHAIN
AO
Account Officer files
UM
San Mateo
AM
Luzon 2
Requests can only advance one level at a time — the Area Manager never sees a filing the Unit Manager has not endorsed.


Recent applications
Open tracker
JA
Joselito Arines LA-2026-0014
Vacation Leave · Nov 29, 2026 – Nov 29, 2026 · 1 day
Pending UM
Filed Nov 6, 2026
JA
Joselito Arines LA-2026-0013
Vacation Leave · Dec 20, 2026 – Dec 21, 2026 · 2 days
Approved
Filed Nov 5, 2026
CM
Charmie Melody Mendoza LA-2026-0011
Vacation Leave · Nov 22, 2026 – Nov 23, 2026 · 2 days
Pending UM
Filed Nov 5, 2026
MA
Mon Alvin Cruz LA-2026-0010
Paternity Leave · Nov 8, 2026 – Nov 12, 2026 · 5 days
Approved
Filed Nov 5, 2026
JA
Joselito Arines LA-2026-0007
Vacation Leave · Nov 8, 2026 – Nov 9, 2026 · 2 days
Approved
Filed Nov 5, 2026


LAR-Sys
CARD MRI RIZAL BANK
Dashboard
For review
Tracker
UM
AG
Aniebel Gay Apelado
Unit Manager
Live status tracker
LA-2026-0014 • San Mateo
Pending UM Recommendation
Vacation Leave
Nov 29, 2026 – Nov 29, 2026 · 1 day · filed Nov 6, 2026, 9:19 AM
SUBMITTED
PENDING UM
3
PENDING AM
4
FINAL
FILED BY
Joselito Arines
UNIT / CLUSTER
San Mateo · Luzon 2
RELIEVER
Mon Alvin Cruz
UNIT MANAGER
Aniebel Gay Apelado
REASON FOR LEAVE
Family Gathering
RELIEVER AVAILABILITY
Mon Alvin Cruz is available
No overlapping leave on record for this period.



LA-2026-0014 — Recommended by your Unit Manager Inbox ×

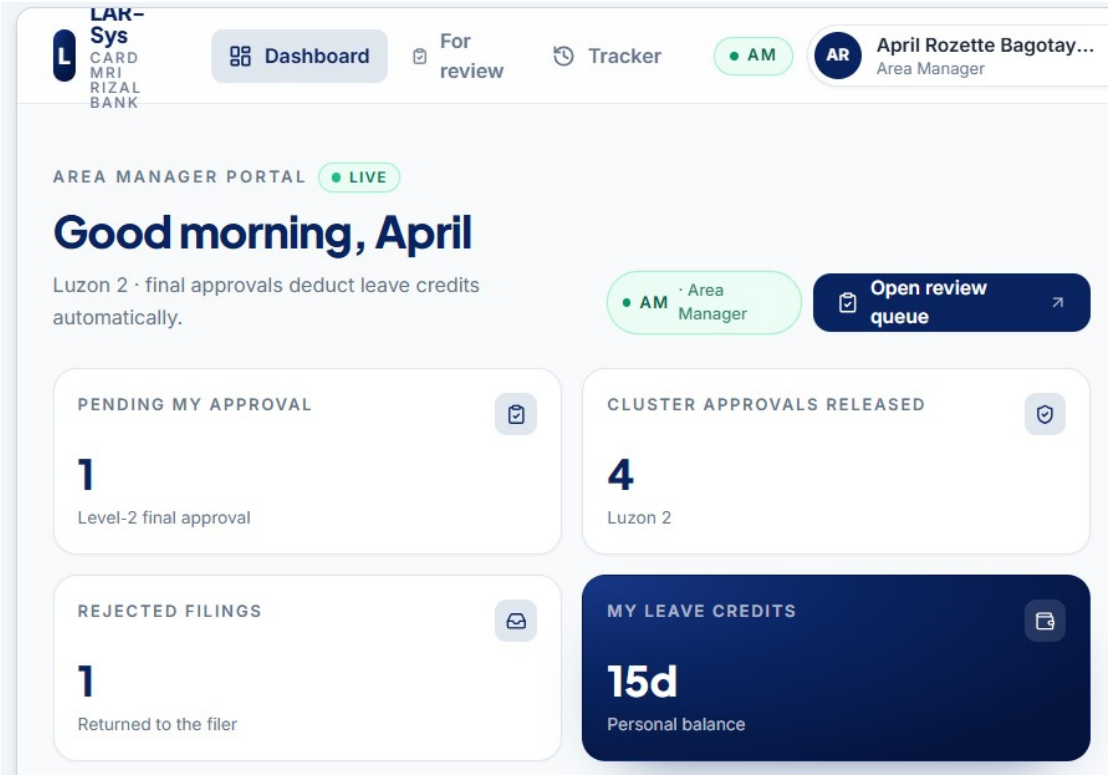
Joselito Arines <joselitoarines31992@gmail.com> 09:25 (0 minutes ago)
to me

Hello Joselito Arines,

Your leave request has been recommended by your Unit Manager, Aniebel Gay Apelado.

Reference: LA-2026-0014
Leave type: Vacation Leave
Dates: 2026-11-30 to 2026-11-30 (1 day)
It now awaits the Area Manager's approval.

— LAR-Sys · CARD MRI Rizal Bank



Leave requests

2 request(s) shown

View all

Pending 2 | Recommended 1 | Approved 4

JA **Joselito Arines** LA-2026-0014
Vacation Leave · Nov 29, 2026 – Nov 29, 2026 · 1 day — Pending AM

CM **Charmie Melody Mendoza** LA-2026-0011
Vacation Leave · Nov 22, 2026 – Nov 23, 2026 · 2 days — Pending UM

LEAVE BALANCE

15 days

Out of a 20-day annual allocation

ROUTING CHAIN

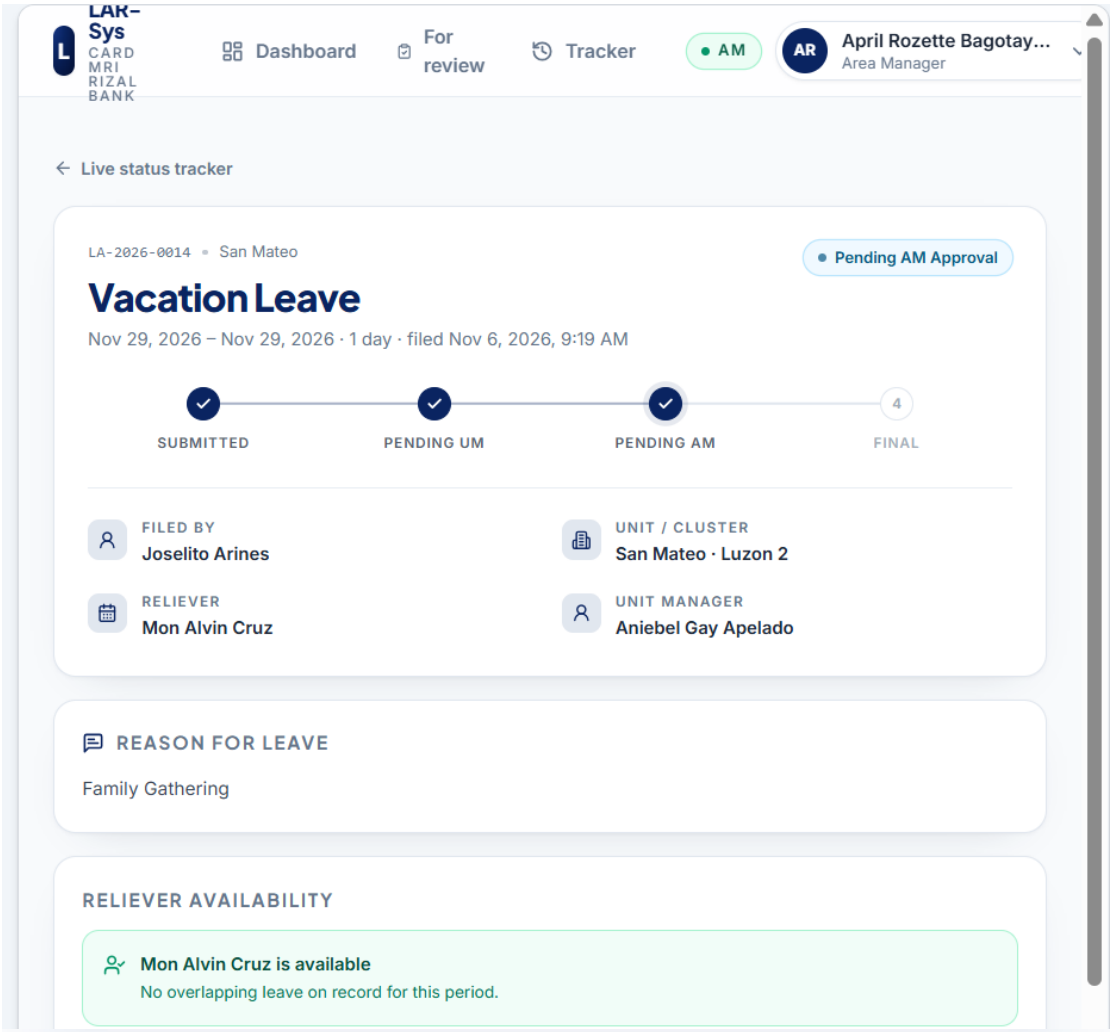
- AO — Account Officer files
- UM — San Mateo
- AM — Luzon 2

Requests can only advance one level at a time — the Area Manager never sees a filing the Unit Manager has not endorsed.

Recent applications

Open tracker

JA	**Joselito Arines** LA-2026-0014 Vacation Leave · Nov 29, 2026 – Nov 29, 2026 · 1 day	Pending AM	Filed Nov 6, 2026
JA	**Joselito Arines** LA-2026-0013 Vacation Leave · Dec 20, 2026 – Dec 21, 2026 · 2 days	Approved	Filed Nov 5, 2026
JM	**Jellow Mercene** LA-2026-0012 Vacation Leave · Dec 13, 2026 – Dec 13, 2026 · 1 day	Cancelled	Filed Nov 5, 2026
CM	**Charmie Melody Mendoza** LA-2026-0011 Vacation Leave · Nov 22, 2026 – Nov 23, 2026 · 2 days	Pending UM	Filed Nov 5, 2026
MA	**Mon Alvin Cruz** LA-2026-0010 Paternity Leave · Nov 8, 2026 – Nov 12, 2026 · 5 days	Approved	Filed Nov 5, 2026



APPROVAL WORKFLOW

Application filed — Nov 6, 2026, 9:19 AM
Joselito Arines · Vacation Leave · 1 day(s)

Unit Manager recommended — Nov 6, 2026, 9:28 AM
Aniebel Gay Apelado
Remarks: Endorsed

Area Manager approved — Nov 6, 2026, 9:28 AM
April Rozette Bagotayao
Remarks: Approved- With Reliever.

WORKFLOW STATUS

Workflow complete — leave approved and credits deducted.

LA-2026-0014 — Approved by your Area Manager Inbox ×

Joselito Arines <joselitoarines31992@gmail.com> 09:28 (0 minutes ag
to me

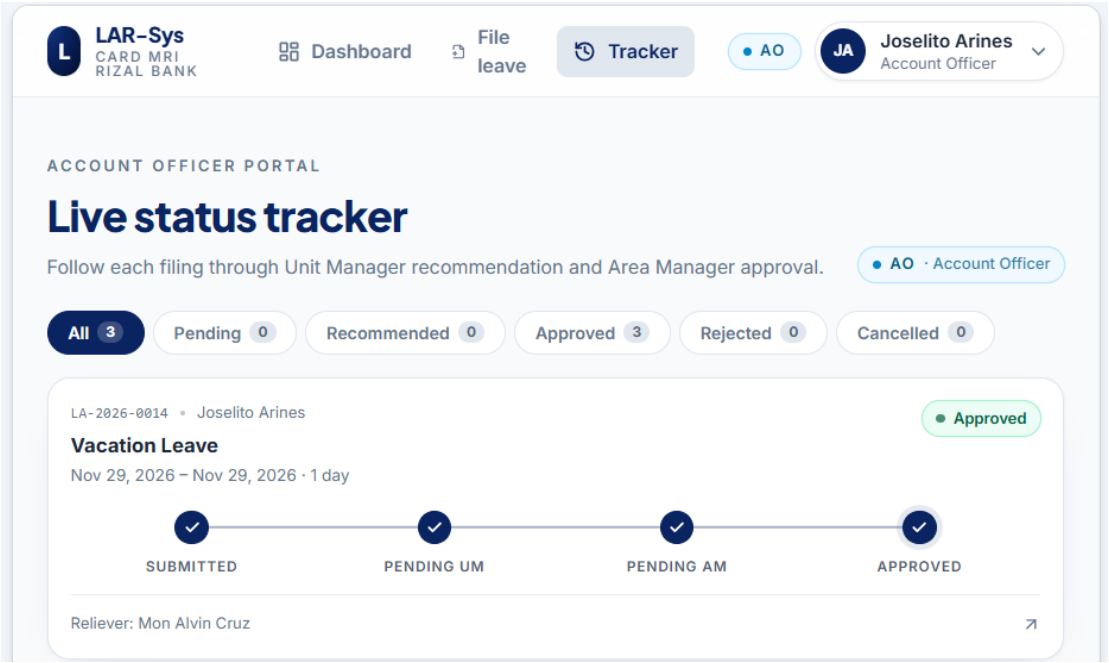
Hello Joselito Arines,

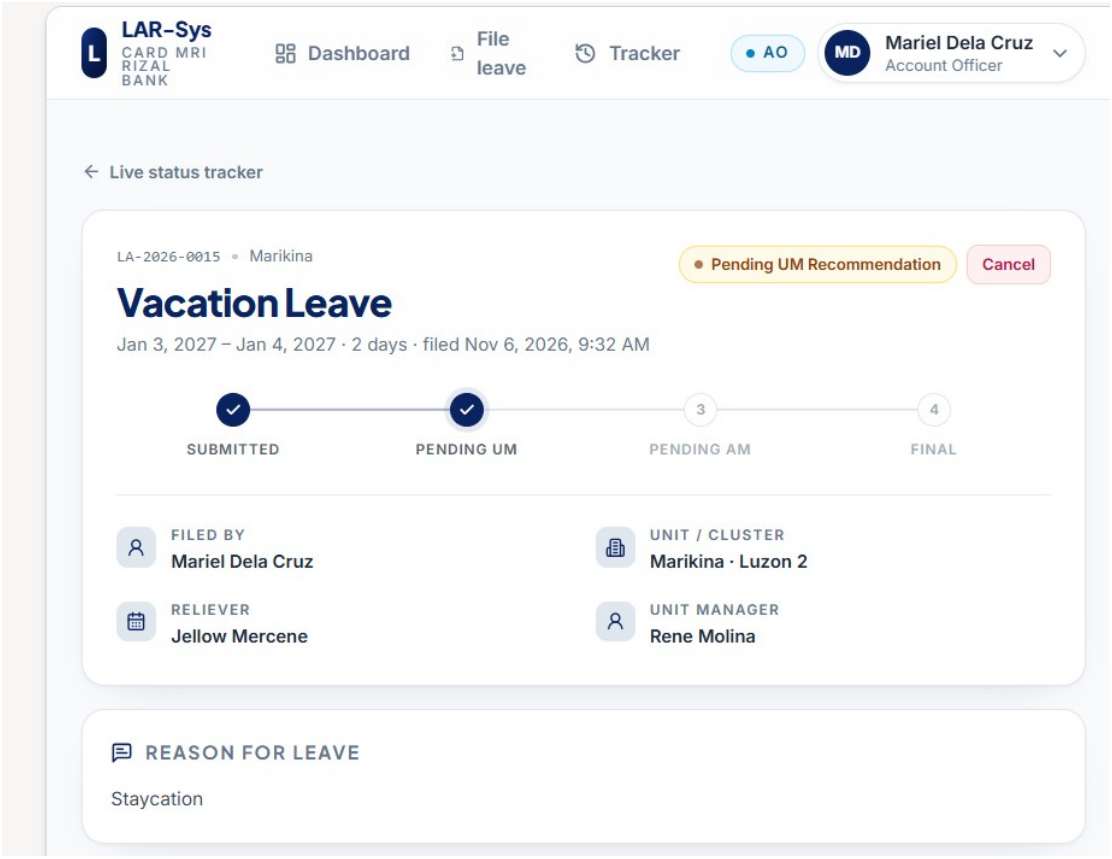
Your leave request has been approved by your Area Manager, April Rozette Bagotayao.

Reference: LA-2026-0014
Leave type: Vacation Leave
Dates: 2026-11-30 to 2026-11-30 (1 day)
Your leave is now fully approved.

— LAR-Sys · CARD MRI Rizal Bank

Reply Forward




LAR-Sys
CARD MRI RIZAL BANK
Dashboard
File leave
Tracker
AO
MD
Mariel Dela Cruz
Account Officer
Live status tracker
LA-2026-0015 • Marikina
Pending UM Recommendation
Cancel
Vacation Leave
Jan 3, 2027 – Jan 4, 2027 · 2 days · filed Nov 6, 2026, 9:32 AM
SUBMITTED
PENDING UM
3
PENDING AM
4
FINAL
FILED BY
Mariel Dela Cruz
UNIT / CLUSTER
Marikina · Luzon 2
RELIEVER
Jellow Mercene
UNIT MANAGER
Rene Molina
REASON FOR LEAVE
Staycation

LEAVE BALANCE

15 days

Out of a 20-day annual allocation

ROUTING CHAIN

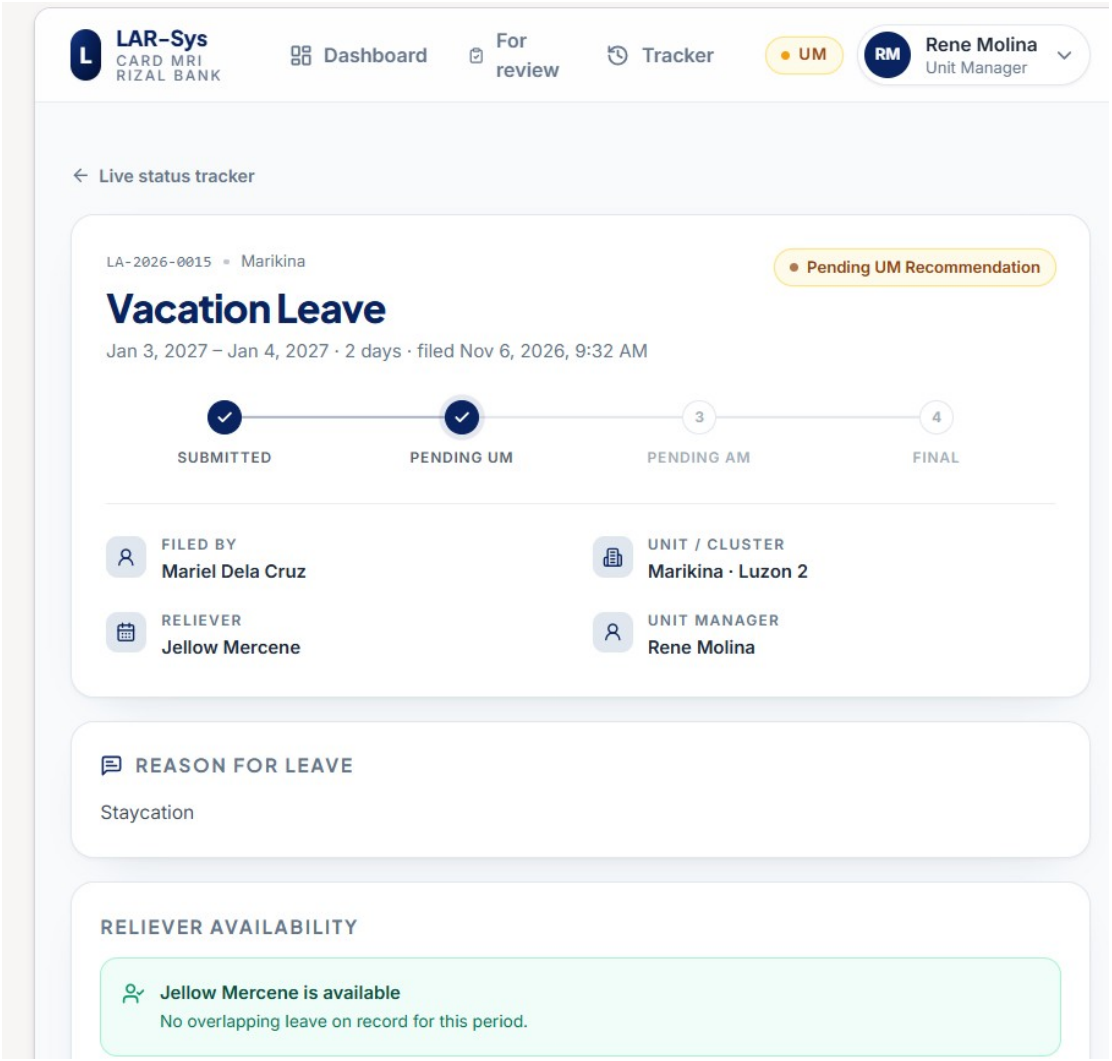
- AO — Account Officer files
- UM — Marikina
- AM — Luzon 2

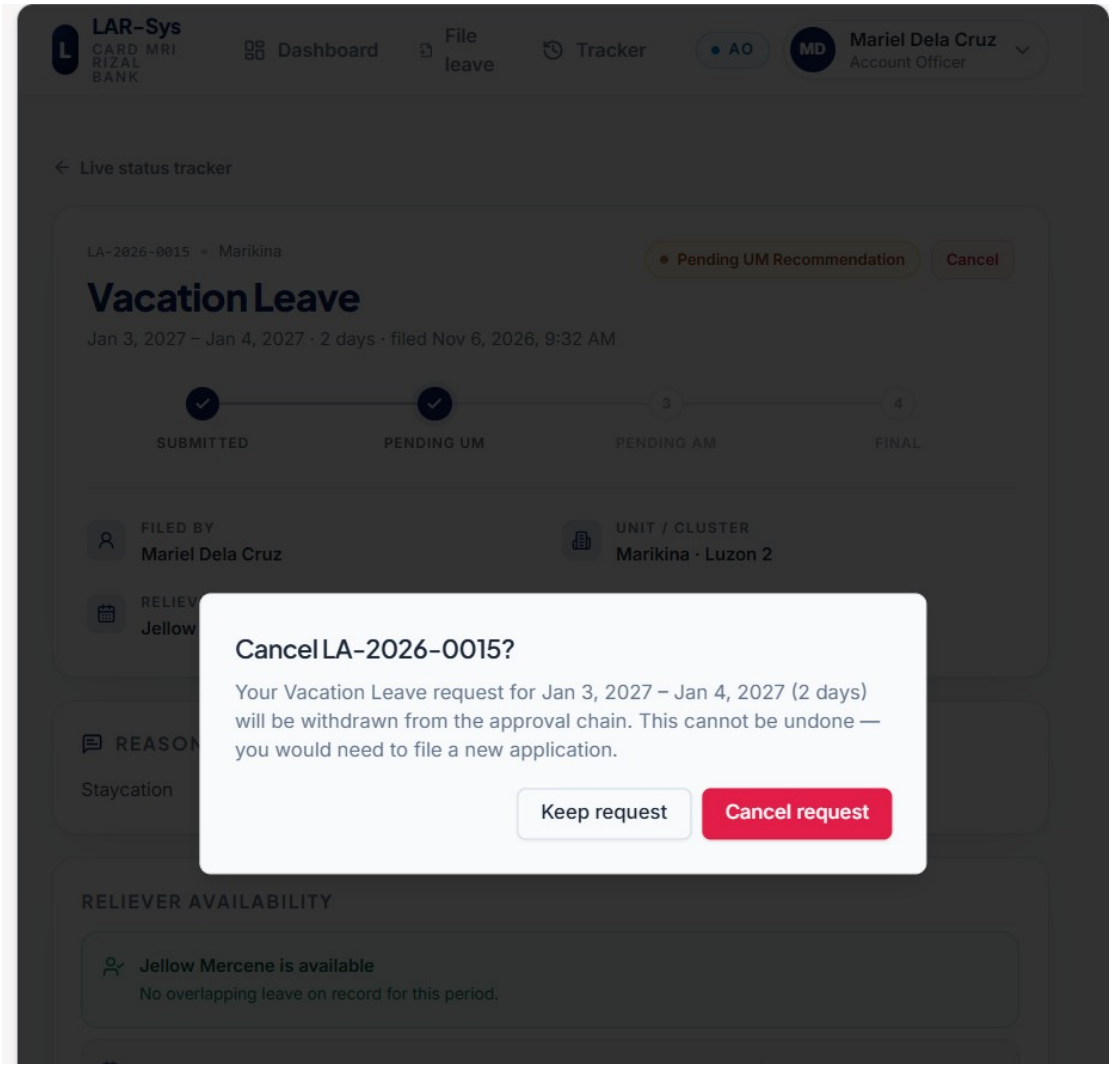
Requests can only advance one level at a time — the Area Manager never sees a filing the Unit Manager has not endorsed.

Recent applications

Open tracker

MD	**Mariel Dela Cruz** LA-2026-0015 Vacation Leave · Jan 3, 2027 – Jan 4, 2027 · 2 days	Cancelled	Filed Nov 6, 2026
JM	**Jellow Mercene** LA-2026-0012 Vacation Leave · Dec 13, 2026 – Dec 13, 2026 · 1 day	Cancelled	Filed Nov 5, 2026
MD	**Mariel Dela Cruz** LA-2026-0009 Vacation Leave · Nov 8, 2026 – Nov 9, 2026 · 2 days	Approved	Filed Nov 5, 2026
MD	**Mariel Dela Cruz** LA-2026-0008 Sick Leave · Nov 3, 2026 – Nov 4, 2026 · 2 days	Rejected	Filed Nov 5, 2026


LAR–Sys
CARD MRI
RIZAL BANK
Dashboard
For review
Tracker
UM
RM
Rene Molina
Unit Manager
Live status tracker
LA-2026-0015 • Marikina
Pending UM Recommendation
Vacation Leave
Jan 3, 2027 – Jan 4, 2027 · 2 days · filed Nov 6, 2026, 9:32 AM
SUBMITTED
PENDING UM
3
PENDING AM
4
FINAL
FILED BY
Mariel Dela Cruz
UNIT / CLUSTER
Marikina · Luzon 2
RELIEVER
Jellow Mercene
UNIT MANAGER
Rene Molina
REASON FOR LEAVE
Staycation
RELIEVER AVAILABILITY
Jellow Mercene is available
No overlapping leave on record for this period.


LAR-Sys
CARD MRI RIZAL BANK
Dashboard
File leave
Tracker
AO
MD
Mariel Dela Cruz
Account Officer
Live status tracker
LA-2026-0015 • Marikina
Pending UM Recommendation
Cancel
Vacation Leave
Jan 3, 2027 – Jan 4, 2027 · 2 days · filed Nov 6, 2026, 9:32 AM
SUBMITTED
PENDING UM
3
PENDING AM
4
FINAL
FILED BY
Mariel Dela Cruz
UNIT / CLUSTER
Marikina · Luzon 2
Jellow
Cancel LA-2026-0015?
Your Vacation Leave request for Jan 3, 2027 – Jan 4, 2027 (2 days) will be withdrawn from the approval chain. This cannot be undone — you would need to file a new application.
Keep request
Cancel request
Staycation
RELIEVER AVAILABILITY
Jellow Mercene is available
No overlapping leave on record for this period.